

Guidelines for Treatment of Atopic Eczema (Atopic Dermatitis) – Part I

Abstract

The evidence for the treatment of atopic eczema (atopic dermatitis, AE) is evaluated according to the Appraisal of Guidelines for Research and Evaluation (AGREE) instrument. The consensus process included a nominal group technique and a DELPHI procedure. Management of AE must account for the individual variability in symptoms and disease course.

Basic therapy centers on regular use of hydrating topical agents and avoidance of specific and non-specific triggering factors. For flare-ups, **anti-inflammatory treatment** with topical glucocorticosteroids or topical calcineurin inhibitors (TCIs) is recommended. TCIs (tacrolimus and pimecrolimus) are increasingly used not only for exacerbations but also for proactive therapy in selected cases, particularly in sensitive areas such as the face and neck.

Topical corticosteroids remain the first-line anti-inflammatory treatment. However, TCIs are preferred in certain anatomical locations due to lower risk of skin atrophy. For severe, refractory cases, **systemic immunosuppressive therapy** may be considered.

Microbial colonization—especially by *Staphylococcus aureus*—and superinfection can trigger flares and may warrant adjunctive antimicrobial treatment. **Phototherapy**, particularly with UVA1 or narrowband UVB (311 nm), is an effective adjuvant option.

Dietary interventions should be implemented only in cases of confirmed food allergy. **Allergen-specific immunotherapy** to aeroallergens may benefit

selected patients. Psychological stress can exacerbate AE, making **psychosomatic counseling** advisable in some cases. Structured patient education programs, such as "**eczema schools**", have demonstrated efficacy.

Pruritus is a central symptom targeted by most therapies, but some patients require additional **antipruritic treatments**.

Introduction

Atopic eczema (AE), also known as atopic dermatitis, eczema, or *Neurodermitis* in German-speaking countries, is a chronic or chronically relapsing, inflammatory, and intensely pruritic skin disorder. It frequently occurs in families with a history of other atopic conditions, such as bronchial asthma or allergic rhinoconjunctivitis.

AE is one of the most prevalent skin diseases worldwide, affecting up to **20% of children** and **1–3% of adults**. It often represents the initial stage of the "atopic march," preceding the development of allergic rhinitis and asthma.

Diagnosis of AE relies on established clinical criteria. There is no pathognomonic laboratory biomarker. While elevated total or allergen-specific IgE levels and positive skin prick tests are common, they are not universal. Patients without IgE sensitization are classified as having **intrinsic (non-IgE-associated) AE**, in contrast to the **extrinsic (IgE-associated) form**. This distinction remains debated but has practical implications for management, particularly regarding allergen avoidance strategies.

Aetiopathophysiology

The development of AE involves a complex interplay of genetic, immunological, and environmental factors:

- **Genetic predisposition:** High heritability is evident, with 80% concordance in monozygotic twins versus 20% in dizygotic twins.
- **Immune dysregulation:** Early disease stages are characterized by a Th2-dominant immune response, leading to increased IgE production.

- **Skin barrier dysfunction:** Impaired barrier function results from abnormalities in lipid composition and structural proteins, including filaggrin mutations and deficiency in protease inhibitors, contributing to dry skin and increased allergen penetration.
- **Microbial dysbiosis:** Abnormal colonization with pathogens such as *Staphylococcus aureus* and *Malassezia furfur*, instead of commensals like *Staphylococcus epidermidis*, increases infection risk and inflammation.
- **Psychosomatic factors:** Stress influences disease activity via autonomic nervous system imbalance and increased release of inflammatory mediators from cells such as eosinophils.

Effective management of AE requires an individualized approach that addresses the specific pathogenic mechanisms present in each patient.

Disease Severity Assessment

After diagnosis, assessment of disease severity is essential. The **Scoring Atopic Dermatitis (SCORAD)** index, developed by the European Task Force on Atopic Dermatitis (ETFAD), is the standard tool for evaluating extent, intensity, and subjective symptoms (pruritus and sleep disturbance).